Tensor computations

Nonnegative H-eigenvalue inheritance from odd-order Hankel tensors

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Challenging because odd-order H-eigenvalue positivity lacks the even-order polynomial-positivity argument; specialist importance lies in transferring spectral certificates within Hankel representations.

Resolution — 2026-09-11

Negative resolution. Matthew J. Colbrook's complete manuscript, Counterexample proposition and proof disproves the universal conjecture with $m = 3$, $q = 2$, $n = 2$ and the common generating vector $h = (2, 0, 1, 0, 2, 0, -1)$. Every real H-eigenvalue of the order-three, dimension-three tensor A is strictly positive, whereas the order-six, dimension-two tensor B has the exact H-eigenpair $(-1, (0, 1))$. The manuscript also proves that A has a real H-eigenpair, so its premise is nonvacuous. The counterexample meets the original odd-lower-order assumptions; it does not contradict the separate even-lower-order theorem or results requiring a positive-semidefinite associated Hankel matrix.

The complete source passed independent Codex-agent proof review. Authorship, AI assistance and verification record. No external human peer review or formal verification is asserted. The original target below is retained verbatim; the difficulty, importance and rating rationale above are historical. This entry no longer contributes to the open count.

Manuscript PDF.

Statement

For every odd $m \geq 3$, integer $q \geq 2$, dimension $n \geq 2$, and vector $h \in \mathbb{R}^{qm(n-1)+1}$, define Hankel tensors A and B with this same generating vector. Their orders and dimensions are

$$\operatorname{order}(A) = m, \quad \dim(A) = q(n-1) + 1, \qquad \operatorname{order}(B) = qm, \quad \dim(B) = n.$$

An order-s, dimension-N Hankel tensor has entry $h_{i_1+\cdots+i_s-s}$, with each index in $\{1, \ldots, N\}$.

For a real order-s tensor C, an H-eigenvalue is a real λ admitting a real nonzero vector x such that, for every i,

$$\sum_{i_2,\ldots,i_s=1}^{N} C_{ii_2\cdots i_s} x_{i_2} \cdots x_{i_s} = \lambda x_i^{s-1}.$$

Conjecture: if A has no negative H-eigenvalues, then B has no negative H-eigenvalues.

Relevance

The question transfers a spectral positivity certificate between different Hankel representations of the same data, relevant to structured tensor eigenvalue computation and polynomial positivity.

References

1. W. Ding, L. Qi, and Y. Wei, *Inheritance properties and sum-of-squares decomposition of Hankel tensors: theory and algorithms*, BIT Numer. Math. 57 (2017), 169–190. DOI; author PDF, final §4, "The third inheritance property of Hankel tensors," concluding conjecture; §2 gives order/dimension conventions.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. Primary preprint, §5, concerning H-eigenvalues and complete Hankel tensors.

Status check — 2026-09-10

Rechecked the Ding–Qi–Wei journal text, final §4, and searched for proofs or counterexamples to the third inheritance property. The source explicitly leaves odd lower order unresolved; its even-lower-order result lies outside the displayed target. Complete or strong Hankel hypotheses would be additional restrictions. No later resolution was located; this remains a bounded historical-source check.